Oblique wall impact

LAMMPS dem04 | Tangential friction impulse and spin | 28 October 2026

PASS	All four runs pass all three official 1% checks at 28.57 us.

Official rebound checks at 28.57 us

Run	vx (m/s)	Err. (%)	vz (m/s)	Err. (%)	omega_y	Err. (%)	1%
CPU dt	1.96264625	0.0017	2.92782788	0.0007	542.750639	0.2702	PASS
GPU dt	1.96264625	0.0017	2.92782788	0.0007	542.750639	0.2702	PASS
CPU dt/2	1.96265101	0.0014	2.92789206	0.0015	542.745838	0.2711	PASS
GPU dt/2	1.96265101	0.0014	2.92789206	0.0015	542.745838	0.2711	PASS

omega_y is in rad/s. Targets: vx = 1.962678784 m/s, vz = 2.927848 m/s, omega_y = 544.221216 rad/s. The official oblique_impact criterion tests each relative error independently against 1% at cumulative step 2857, the second segment end. Golden-state comparisons are a separate diagnostic.

Inputs and direct contact mapping

Quantity	Official input / FunDEM configuration
Sphere / initial state	Diameter 0.005 m; density 2700 kg/m^3; center (0, 0, 0.002550) m; v = (2.5069, 0, -2.9876) m/s; zero spin
Wall / other forces	Fixed plane z = 0; gravity, rolling/torsional friction and global damping = 0; no boundary crossing
Normal / tangential	Hooke kn = 2e7 N/m; linear_history kt = 5.7e6 N/m; e = 0.98; sliding mu = 0.092
Explicit damping	cn = 2 beta sqrt(m kn) = 0.7645958085 N s/m; ct = cr = ctorsion = 0; beta = -ln(e)/sqrt(pi^2 + ln(e)^2)
Elastic schema fields	Explicit linear stiffness enabled; E = 10000 Pa and nu = 0.33 are inactive placeholders, not official values or fitted inputs.
Time / saved output	dt = 10 / 5 ns; 3757 / 7514 steps; 37.57 us endpoint; every integration step retained in native results and CSV

Friction impulse and angular momentum

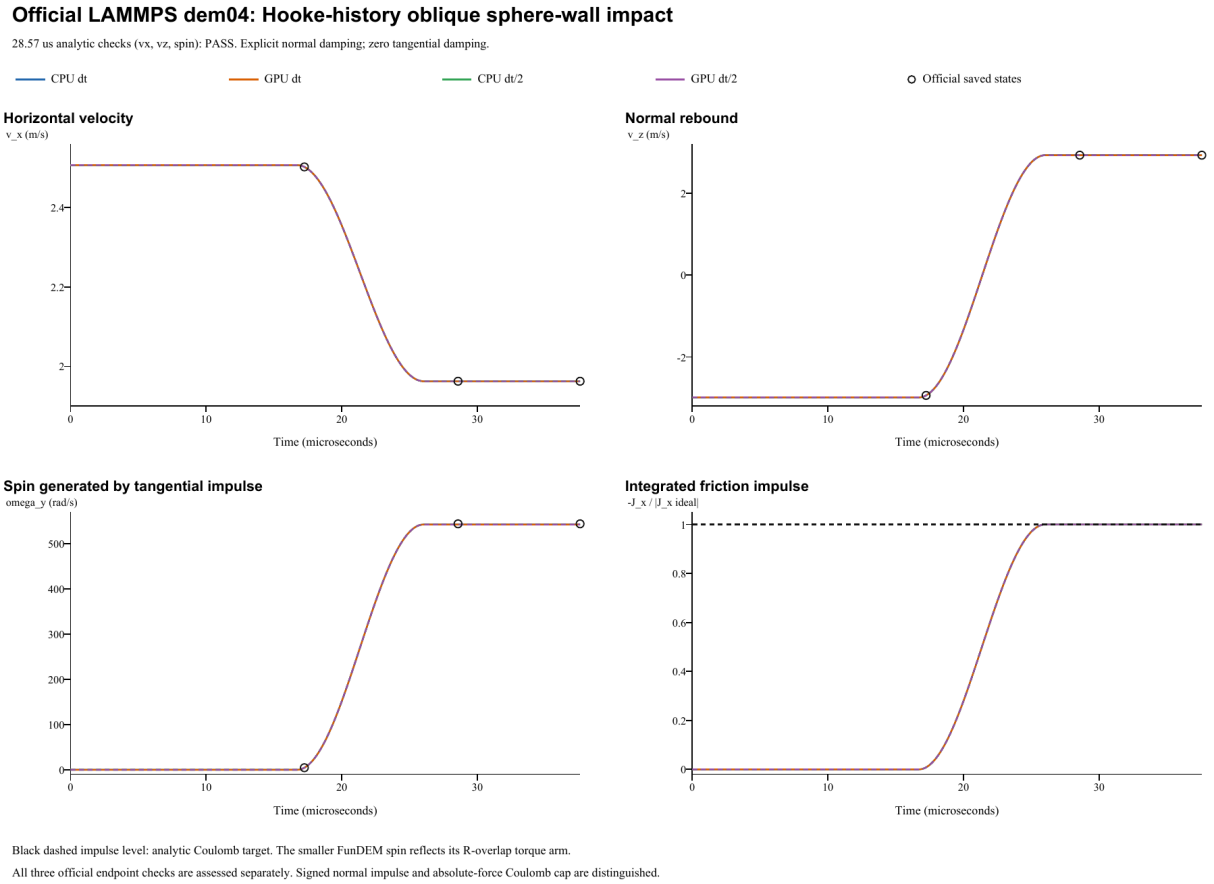
Jn = m (1+e) vz_in = 0.00104534594885 N s.
Jt = -mu Jn = -9.61718272939e-5 N s.
Angular impulse = -R Jt = 2.40429568235e-7 N m s; I = (2/5) m R^2.

Explicit cn maps prescribed restitution while preserving official ct = 0. The damping override is enabled; this is not a test of automatic Hooke normal or tangential damping. The three continuous official segments are one uninterrupted native event.

Reference provenance

Official YAML and golden states; analytic checker. Commit f9720ce2d176d615495d1d6c31349b1f71be1c73; YAML Git blob 6ab54dcfe7fcef2d4dcc16eb430a29ba8ebe8f3a. LAMMPS was not rerun locally. YAML epsilon 1e-10 is not a universal cross-code tolerance. No solver or input parameter was changed to force agreement.

Curves and retained evidence



Capped abs(Ft)/abs(Fn) = 0.092; slip stays positive (minimum 0.6058012 m/s). Five base-step samples are uncapped history transients. Impulse-balance residuals are at most 2.82e-18 kg m/s and 1.96e-21 N m s. CPU/GPU spin differs by at most 1.71e-13 rad/s. Halving dt changes spin by 0.00480095 rad/s; the 0.27% gap persists. The base-dt signed Jn / Jt are 0.00104534239 / -9.61775762e-05 N s.

FunDEM uses a = R - delta for ordinary friction torque; LAMMPS uses R. The base-step weighted arm is 0.002493095551 m. Signed Fn sets momentum while abs(Fn) caps friction; the weak tensile tail is retained. The official targets remain unchanged.

Complete native results are retained

CPU dt: run.json | GPU dt: run.json | CPU dt/2: run.json | GPU dt/2: run.json

Base dt: 3757 steps / 3758 frames per run; half dt: 7514 steps / 7515 frames per run. Every numerical step and all 22,546 native frames are retained.

Run manifest | Verified analysis | Checkpoint CSV | Contact audit | Full native archive ZIP. All native frames/contacts, projects, audits, backend logs and every-step CSVs are retained.

Execution and scope

Real SimulationSession runs use linked 0.4.3 CPU/CUDA native libraries without solver changes. The three official 1% criteria determine pass/fail. GUI executable byte identity was not established.

This case checks oblique-impact friction impulse and spin. Rolling/torsional resistance are zero. Gross sliding does not fully test sticking, history reversal, stick-slip cycles, nonzero tangential damping or many-particle flows.